MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 813

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, beam power

See figure 1

Mounting position: Vertical, base up or down; horizontal, anode and vertical plane

Weight: 8-ounces (226.8 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec1	Ec2	Ec3	Ib	Ic2	Pg2	Pp	Pi
Unit:	MHz	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	W
Maximum:	30	10.5	2,000	-300	400	---	180	25	22	100	360
Minimum:	---	9.5	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	10 Vdc	2,000	---	400	0	---	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	4.70	5.30	Adc
1266	Total grid current	1	Eb/Ib = 50 mAdc	0.65	II	Ic1	0	-10.0	µAdc
1256	Electrode current (anode)	-	Ec = 35 Vdc	0.65	II	Ib	35	65	mAdc
1236	Power oscillation (1)	-	Rg = 10,000 ohms; Ib = 180 mAdc; Ic = 12 mAdc; F = 15 MHz	0.65	II	Po	220	---	W
1256	Electrode current (screen)	-	Power oscillation (1)	0.65	II	Ic2	25	55	mAdc
1231	Emission by oscillation	-	Power oscillation (1)	0.65	II	Ef	---	9.0	Vdc
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Eb = Ec2 = 250 Vdc; Ec = -20 Vdc; Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1236	Internal insulation	-	F = 15 MHz	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-	Without shield	---	---	Cgp	---	0.25	pF
						Cin	13.0	19.6	pF
						Cout	10.5	17.5	pF
	Quality conformance inspection, part 3								
---	Life test	-	Group C; Pp = 100 W; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Emission by oscillation	-		---	---	Ef	---	9.5	Vdc
(D) 1236	Power oscillation (2)	2	Rg = 10,000 ohms; Ib = 180 mAdc; Ic = 25 mAdc (max); F = 30 MHz	---	---	Po	180	---	W

NOTES:

1. This test is to be the first test performed at the conclusion of the holding period.

(D) 2. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection (QCI), part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

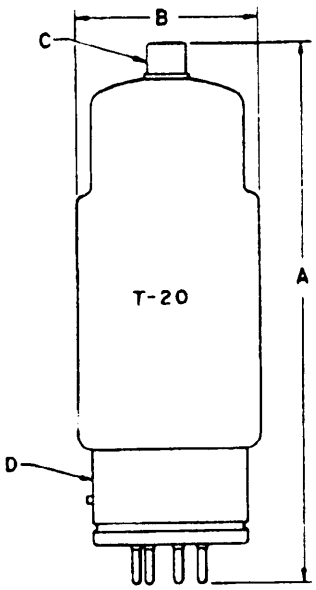
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2923)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	7.50 (190.50)
B	---	2.63 (66.80)
(D) Quality conformance inspection, part 3 (see note)		
C	Cap: Medium C1-5 (EIA)	
D	Base: A7-17 (EIA)	

PIN CONNECTIONS	
Pin No.	Element
1 and 7	h
2	nc
3	g2
4	g1
5	g3
6	nc
cap	a

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 813.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVY DEPARTMENT
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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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